MASTER REFERENCE · CONSOLIDATED FDE SUITE

The StatusNeo Agentic FDE Practice

Charter ·Operating Model ·Lifecycle ·Tooling ·Security ·Architecture ·AEC Use Case

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What this is. A single master reference consolidating the Forward Deployed Engineering practice: the strategic thesis, the 80:20 (AI:Human) operating model, the agentic lifecycle, the multi-cloud tooling directory, the 2026 security and CI/CD standards, the Modutecture AEC enterprise architecture, the contextual-construction use case, and the Kaiser healthcare correctness gate — wrapped in the StatusNeo four-loop GCC execution engine.

Who it's for. The FDE practice lead, the StatusNeo sponsor, and the Modutecture technical stakeholders. This is a working charter-and-scoping reference, not a delivery commitment.

READ THIS FIRST — SOURCING & VERIFICATION

This document consolidates June-2026 source briefings (Modutecture, StatusNeo GCC, the multi-cloud ecosystem). It names specific products, metrics, and dates — AgentCore, Entra Agent IDs, the 7-layer architecture, FGI 2026, 14-day TTFV, 85% margin. **Treat them as the target operating model, pending verification — not validated results.** Durable: the structure (loops, 80:20, architecture). Perishable: the named tools, metrics, and dates.

Version: Master v1 · consolidated June 2026 · companion documents indexed at the end

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I

The Thesis

Why the Forward Deployed Engineering practice — and why now

Every enterprise wanting AI transformation faces the same problem: the gap between a brilliant framework and a landed, trusted production system. StatusNeo's frameworks and offshore scale are strong; what is missing is the **onshore tip of the spear** that lands those frameworks inside the customer, de-risks the handoff, and unlocks the regulated, air-gapped enterprises (healthcare, BFSI, defense) that structurally will not buy pure-offshore delivery.

The Elite Pod Commander model

The 2026 FDE has evolved from a custom coder into an **Elite Pod Commander** who orchestrates an autonomous AI swarm. Per the source, this lets a single FDE manage complex pipelines that previously required 5–10 manual engineers — replacing a heavy-headcount consulting team with one commander backed by standardized state machines.

The old model	The Agentic FDE model (per source)
A 10-person consulting team, multi-year headcount commitment.	One Elite Pod Commander + an autonomous AI swarm.
Custom glue code, manual integration, slow trust-building.	Spec-Driven Development, localized compute, a 14-day velocity loop.
Time-to-first-value in 3–6 months.	Time-to-first-value targeted at 14 days (per source, to verify).

Land-and-expand: FDE lands, GCC scales

The FDE is a **mandatory precursor**, not a competing silo. The Pod Commander proves architectural trust in a short, high-intensity engagement (the source cites 14 days); once trusted, the offshore StatusNeo GCC scales delivery behind them. The FDE earns the relationship; the GCC scales the volume. This is the economic engine: bypass the friction of multi-year headcount commitments by proving value first.

THE GOVERNING PRINCIPLE (CARRIED THROUGHOUT)

[redacted]

II

The 80:20 Operating Model

Automating the FDE lifecycle: 80% agentic AI, 20% human-in-the-loop

The FDE is now a two-part entity: the **Agentic FDE** (the 80% — velocity, breadth, repetition) and the **Human FDE / Pod Commander** (the 20% — judgment, trust, architecture, governance). The 80:20 is a leverage ratio, not a uniform rule: the human owns the front (intent + architecture) and the end (the ship gate); the swarm owns the dense middle.

AI 80%	HITL 20%

The Agentic FDE — the 80% (the swarm roles)

Agent role	Owns	Stage
Spec Architect	Translates human intent + spec into a machine-executable plan (Gherkin/SDD).	1–2
Systems Engineer	Writes the code (memory-safe Rust per source), builds the binary.	3
Adversarial SDET	Generates and runs tests; stress-tests its own output for 100% coverage.	4
Security Sentinel	Runs SAST, dependency scans, compliance checks as agent tools.	5
DevOps Overseer	Deploys via zero-trust templates and secure sandboxes within guardrails.	6
Reliability Agent	Monitors production, maps stack traces, cuts auto-hotfix PRs.	7
Platform Architect	Abstracts reusable components from the Spoke back up to the Hub.	cross-cutting

The Human FDE — the 20% (judgment-owned)

Human responsibility	Why it can't be automated
Diagnosis & architecture	Reading the client's real problem requires judgment the swarm has no context for.
Client trust & evangelism	A CISO trusts a person in the room, not a PR bot. Change leadership is human work.
Spec authorship & intent	The swarm builds what the spec says; a wrong spec fails at machine speed.
Governance & the ship gate	In regulated domains a human is accountable for the irreversible call.

The RACI — accountability never automates

The pattern that matters: the **human is Accountable for every stage** even where the AI is Responsible for the work. (Full per-stage RACI in the companion *FDE 80:20 Roles & Lifecycle* document.)

III

The Agentic FDE Lifecycle

The 7-stage SDLC, expressed as the four-loop GCC operating system

Autonomous agents execute a closed-loop lifecycle — pulling tasks from a backlog and pushing verified code to production. Production telemetry feeds back into planning, so the system self-heals and re-plans without a human restart.

The closed-loop agentic SDLC

```
[1. Context & Plan] ──> [2. Architecture] ──> [3. Implementation] ──> [4. Agentic QA]
         ▲                                                                   │
         └──────────── [7. Monitor & Self-Heal] <── [6. CD] <── [5. Security] ◄┘
```

The four-loop GCC expression

StatusNeo expresses this lifecycle as a four-loop operating system aligned to the LoopManifesto's **Observe → Learn → Decide → Act** cycle. Every GCC is not a static “center” but a self-learning ecosystem.

Loop	Cycle phase	Focus	Lifecycle stages
Clarity	Observe	Product & experience — discovery, strategy, AI product design.	1–2
Build	Learn	Engineering — product, data & AI, platform automation.	3
Velocity	Decide	Automation — DevSecOps, SRE, CloudOps, test automation.	4–6
Improvement	Act	Governance & scale — compliance, observability, maturity index.	7

Two topologies, one codebase

Dimension	Embedded POD	Remote POD
Effective split	~70:30 (AI:Human) — human reviews more.	~90:10 (AI:Human) — automated gates.
The ship gate	Human approves the merge (Slack/GitHub).	Automated policy/eval gate, human-defined.
Best for	Legacy apps, core features, regulated work.	Greenfield, ephemeral micro-services.
Trust posture	“Super-powered intern” earning trust.	Sovereign service, trust already earned.

(Full blueprint, playbook, cookbook, and runbook in the companion *Agentic FDE Lifecycle* document.)

IV

The Multi-Cloud Tooling Directory

Cloud-agnostic core, cloud-native bindings — mapped to topology

Remain cloud-agnostic at the core while natively integrating each provider's agentic services. One orchestration codebase; four interchangeable execution substrates. *[Product names and dates per source, pending verification.]*

Pillar	Generic / OSS	AWS	GCP	Azure
Orchestration	LangGraph, CrewAI, AutoGen	Bedrock AgentCore / Flows	Gemini Enterprise Agent Platform	Azure AI Agent Service
Compute Sandbox	E2B, Docker, Fly.io	Lambda / ECS; AgentCore microVMs	Cloud Run v2 (gVisor)	Container Apps Dynamic Sessions
Context / Memory	Greptile, Qdrant	AWS Context Service	Agent Engine “Memory Banks”	Cosmos DB Agent Memory
Code Models	DeepSeek-Coder, Llama-3	Anthropic Claude	Gemini 1.5 / 3	Azure OpenAI GPT / o-series
Compute (per src)	Local RTX 4090 nodes	EC2 G7 (Blackwell)	TPU 8t / 8i	—

Embedded vs. Remote crosswalk

Component	Embedded (Human-in-the-Loop)	Remote (Autonomous)
Persona	Individual developers, team leads.	DevOps engineers, SREs, product owners.
Interface	IDE (VS Code, Cursor), Slack, GitHub PRs.	API gateway, event bus (Kafka/SQS).
Core tools	Kiro, Foundry VS Code, Strands SDK.	Bedrock AgentCore, GCP Agent Runtime.
Acceptance	Human code review / approval gate.	Automated evaluations / canary deploys.

V

Agentic Security & Guardrails

Perimeter-level security: guardrails in the gateway, not the agent code

In 2026 the industry has shifted to **perimeter-level security**, where guardrails live in the cloud gateway rather than inside the agent's application code — so they cannot be bypassed by agent-logic failures. *[Vendor features and dates per source, pending verification.]*

Multi-cloud security feature comparison (per source)

Concern	AWS Bedrock AgentCore	Azure AI Foundry	GCP Gemini Platform
Prompt injection	Gateway-native: blocks injection before the model.	Cross-prompt classifier scans tools/email triggers.	Adaptive filters in a Gemini-native SOC.
PII & data	InvokeGuardrail API — character-offset redaction.	Microsoft Purview — end-to-end DLP.	Zero Data Retention commitment.
Identity	IAM Agent Roles — scoped tool permissions.	Entra Agent ID — eliminates “shadow agents”.	Workspace Studio — integrated RBAC.
Monitoring	AgentCore Observability — logs every policy eval.	Agent Registry — centralized behavior audit.	A2A Audit — cross-cloud call tracking.

2026 best practices for engineers

1. **Strict input/output validation.** Production enforces 100% validation for critical traffic (customer data, payments).
2. **Decoupled policy enforcement.** Use cloud-native gateways so guardrails can't be bypassed by agent-logic failures.
3. **Deterministic pre-checks.** Keep PII and injection checks fast and deterministic (regex) before slow LLM evaluators.
4. **Agent identity.** Treat agents as autonomous identities with their own RBAC, not extensions of user sessions.

VI

CI/CD for Remote PODs

The 2026 deployment standard: mandatory security and evaluation gates

Deploying a Remote FDE POD (an autonomous agent service) requires a pipeline with mandatory security and evaluation gates before production. Per source, **SLSA Level 3** is now the minimum for any agent touching production infrastructure. (Implementation-level pipeline config is held for the build phase; this is the gate model.)

The pipeline gate model

Stage	Gate	What it enforces
1. Security audit	Secret scanning + config validation	Entropy analysis for LLM keys (TruffleHog); validate Bedrock/Foundry guardrail policy files against schema.
2. Evaluation gate	Batch evals + groundedness	Run baseline evaluations to catch reasoning regressions; fail if groundedness < 0.95 (2026 standard).
3. Deploy	Isolated runtime + agent identity	Deploy to isolated compute (Container Apps Dynamic Sessions); register the agent identity (Entra Agent ID).

Essential deployment gates

- **Secret management:** HashiCorp Vault or AWS Secrets Manager with OIDC for just-in-time credentials.
- **Supply-chain integrity:** SLSA Level 3 minimum for production-touching agents.
- **Automated rollbacks:** A/B traffic splitting with automatic rollback if guardrail trigger rates spike.
- **SLA compliance:** FDE contracts typically mandate "time-to-first-integration" ≤ 14 days and production within 90 days (per source).

VII

The Modutecture AEC Architecture

From static models to a Governed Agentic Engine — 7 layers, 4 pillars

The Modutecture Enterprise Architecture (per source) transforms static architectural models into **Governed Agentic Engines**. The Digital Twin is the authoritative source of truth; a multi-agent layer handles reasoning, validation, and real-time operations. The foundation is domain-agnostic AEC, specialized for the Kaiser Permanente healthcare vertical.

The 7-layer production stack

Layer	Component	Function (per source)
1. Experience	Interchangeable renderers	Unity / UE5 / Web act as thin-client lenses; the renderer "owns nothing".
2. Domain Context	Vertical packs	Healthcare ontology (med-gas, clinical room types) and tuned LoRAs.
3. AEC Context	Intelligence foundation	Geometry, spatial semantics, collision prediction (GraphRAG).
4. Agentic AI	Orchestration	Brain proposes, gate disposes: LangGraph + MCP tools for clinical workflows.
5. Twin & Knowledge	Core memory	Event-sourced twin (source of truth) + semantic memory (Neo4j).
6. Platform Services	Seams & contracts	GraphQL + REST APIs with a CQRS command/query split.
7. Data & Processing	Polyglot persistence	Redpanda event backbone, Databricks, SQL/NoSQL storage.

The 4 governance pillars (cross-cutting)

Pillar	What it ensures
Business & stakeholder context	Outcome-based metrics like "time-to-compliant-design."
Security, identity & multitenancy	HIPAA/PHI safeguards and RBAC via Entra Agent IDs.
Governance, safety & compliance	The Deterministic Gate with binding authority over building codes (FGI, ADA, IBC).
DevOps, observability & quality	SLSA-compliant CI/CD and agent-behavior metrics (OpenTelemetry).

Three implementation insights

- **The renderer is a lens.** Unity/UE5 are an "execution lens" (GPU); the agentic engine owns the logic. (Open question: where authoritative geometry lives — see Part XI.)
- **Correctness-by-construction.** The deterministic gate prevents illegal placements (e.g., healthcare-compliance errors) in real time — an "AI safety net."
- **A2A interoperability.** A2A Protocol + MCP let the system plug into AEC standards like IFC and BCF.

VIII

The AEC Use Case — Contextual Construction

The four loops applied to Modutecture, and the FDE 80:20 case study

The use case: embed a Pod Commander + AI swarm in Modutecture's "Continuous Contextual Construction" platform; automate 80% of the twin pipeline (federation, grounding, generation, testing, verification) and reserve the 20% (architecture, compliance, clinical safety sign-off) for the human; prove it on one clinical account through a deterministic correctness gate; then scale into a StatusNeo GCC.

The 14-day velocity loop (the FDE 80:20 case study, per source)

Phase	Autonomous swarm (80%)	Pod Commander (20%)	Tools (per src)
Discovery (Days 1–3)	SDD compilation: maps stakeholder needs to strict Gherkin specs.	C-suite alignment; defines the "why."	Perspective AI
Engineering (Days 4–10)	Swarm generates memory-safe Rust binaries.	Architectural oversight; HIR < 2 commits / 1k LOC.	Agentic Workbench™
Testing & Sync (Days 4–10)	Adversarial SDET → 100% coverage before human review.	Resolves BIM-vs-clinical clashes.	OpenSpace, TestCraft™
Deployment (Days 11–13)	DevOps Agent deploys via zero-trust templates.	Scopes GCC scale-out; abstracts IP to the Hub.	NeoLens™

The four loops, applied

Loop	Split	What happens in the Modutecture use case
Clarity	**30:70**	Perspective AI runs discovery; ingests codes + BIM into GraphRAG; human frames the problem and authors intent.
Build	**90:10**	Swarm does BIM federation + twin-logic generation via SDD; human makes architecture calls.
Velocity	**80:20**	TestCraft™ + Adversarial SDET run the 39-test gate; OpenSpace verifies site-to-twin; human holds the ship gate.
Improvement	**60:40**	NeoLens™ monitors the live twin and drift; human owns governance and decides where to scale.

FDE performance benchmarks (the StatusNeo standard, per source)

- **Time-to-first-value:** reduced from 3–6 months to 14 days.
- **Gross margin:** targeted 85%+ by scaling compute (localized RTX 4090 nodes) rather than headcount.
- **Operating economics:** one Commander per client, offsetting cost to compute instances.

IX

The Kaiser Correctness Gate

Deterministic safety for healthcare AEC: FGI 2026, NFPA 99, HIPAA

In a clinical environment, the agentic speed must never compromise integrity. The **Deterministic Gate** has binding authority over building codes — an artifact passes only on a compile + a compliance suite + a human safety sign-off, never on an agent's say-so. The 2026 healthcare-compliance landscape has shifted toward **deterministic safety**, where mandatory requirements are decoupled from supplemental guidance to enable automated verification.

A NOTE ON THE REGULATORY CLAIMS

FGI Guidelines, NFPA 99, ADA, IBC, and HIPAA are real, established frameworks, and healthcare codes do update on multi-year cycles. The specific 2026 editions and dates below are per source — confirm the exact current editions and effective dates with the client's compliance team before they enter a binding gate.

What the gate must enforce (per source)

Requirement	Standard	Gate behavior
Med-gas systems	NFPA 99 (into the Joint Commission's new “Physical Environment” chapter)	Validate med-gas placement and clinical-room requirements automatically.
Patient privacy	HIPAA — incl. SUD record updates (Feb 16, 2026 per source)	Enforce PHI safeguards and privacy-by-design in the artifact.
Design codes	FGI 2026, ADA, IBC	Prevent illegal placements in real time (correctness-by-construction).

The deterministic gate is the 80:20 made safe for healthcare: **the AI proposes the contextual artifact; the human disposes on whether it is safe to build.**

X

The GCC Execution Engine

From offshore center to self-improving ecosystem — the 90-day path

The StatusNeo GCC Blueprint transforms offshore centers from “shared services” into intelligent systems that improve themselves, using the Agentic FDE as the mandatory precursor that proves trust before the GCC scales.

The 90-day execution roadmap

Timeline	Milestone	Key outcomes (per source)
Day 1–30	The Forge	Define core MCP toolsets, establish local compute nodes, map telemetry to dashboards.
Day 31–60	Proving Grounds	Run 14-day Agentic FDE pilots for beta clients; tune the Human Intervention Rate.
Day 61–90	Global Launch	Publish sanitized case studies; scale out via active product pods.

The Enterprise OS Fabric™ (6 layers, per source)

Component	Purpose	Function in the FDE workflow
Agentic Workbench™	Governed AI sandboxes	Local compute (RTX 4090 nodes) for the FDE AI swarm.
NeoLens™	Observability & AIOps	Real-time release velocity + 14-day pilot success metrics.
TestCraft™	Quality framework	Adversarial SDET loops for zero-trust testing, 100% coverage.
QueryButler™	Knowledge & data access	NL interface for the swarm to query the BIM source of truth.
NeoArkitect™	Golden paths	Zero-trust templates for infrastructure deployment.
Enterprise OS Fabric™	SDLC blueprint	Connects the Agentic FDE pilot to the long-term GCC scale-out.

Measured outcomes (2026 benchmarks, per source)

- **+40–60% faster release velocity** (via Spec-Driven Development).
- **−50% mean-time-to-recovery** (via autonomous AIOps monitoring).
- **Speed to value:** functional product pods live in ≤ 90 days.

XI

Roadmap, Benchmarks & Open Questions

What to resolve in the charter — and the companion documents

The open scoping questions (Day-1 priorities)

1. **Geometry ownership.** Does authoritative Moducule geometry live in the Unity/BIM lens or the twin backend? Decisive test: can a non-Unity client reconstruct a Moducule from backend data alone?
2. **The 39-test gate.** What is the real composition of the correctness gate for the Kaiser/clinical context, and who owns its definition — Modutecture, the client's compliance team, or the FDE?
3. **Compute substrate.** Local RTX 4090 nodes vs. cloud G7 for the Adversarial SDET's fuzzing load, given air-gapped/data-residency constraints?
4. **Rules authority.** Where does rules authority split between Unity-side behaviors and the batch (Databricks) pipeline?
5. **Tool build/buy/verify.** Which named tools are real and licensed today vs. aspirational — Perspective AI, OpenSpace, and the StatusNeo ™ accelerators all need a status before they enter scope.

The honesty discipline (non-negotiable)

Throughout this suite, claims are labeled by confidence and every June-2026 product/metric/date is flagged pending verification. The Modutecture engagement is framed as a **validated demonstration, not a closed-and-scaled account**. Protect that precision — it is the integrity signal that wins regulated rooms.

Companion documents (the full suite)

#	Document	What it holds
01	FDE Charter Proposal	The strategic case + change-leadership + the Copilot×Twin toolbox.
02–03	Loop Cookbook + Landing Runbook	The methodology and the operational field manual.
05	Agentic FDE Lifecycle	The 7-stage SDLC: blueprint, playbook, cookbook, runbook.
06	FDE 80:20 Roles & Lifecycle	The role split, per-stage ratios, and the full RACI.
07	Multi-Cloud Tooling Directory	Generic/AWS/GCP/Azure, Embedded vs. Remote.
08	AEC/Modutecture Use Case	Contextual construction across the four loops.
—	PR-FAQ Playbook + Deck	The Amazon-style narrative and the board deck.

THE WHOLE THING, IN ONE LINE

A single Elite Pod Commander, backed by a governed AI swarm, automates 80% of the AEC twin lifecycle and reserves the judgment-bound 20% — lands one regulated account through a deterministic correctness gate, then scales it into a self-improving StatusNeo GCC. The AI proposes; the human disposes; the gate is the point.